Hours Worked and Development: Comment

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Gottlieb (2026) reproduces the main findings of our article "Global Working Hours" (*Quarterly Journal of Economics*, 2026) and conducts a number of additional analyses on our database. We are very grateful to the author for his careful examination of our database and programs. We posted them publicly precisely so that other researchers would study the database and help further improve it. In this comment, we discuss the main points raised in Gottlieb (2026) and how we will update the database in light of his findings.

1) Hours and GDP Years Gottlieb (2026) rightfully highlights that data on hours worked and GDP per capita are not taken from the same years in our cross-sectional database. We always take hours worked from the most recent year available (excluding COVID years), while GDP per adult and populations are for 2023.[1] We did this because the spirit of our cross-sectional analysis is to study countries around the world at the same point in time. Our implicit assumption is that hours worked from the most recent labor survey data are close enough to hours worked in 2023. This assumption can be justified on the grounds that our analysis shows little change in hours of work overtime, particularly for prime-age adults (the vast majority of adults). However, this basic assumption could be refined by projecting hours worked to 2023 using our data.

Gottlieb is also correct that mapping hours worked onto GDP and populations from the same years slightly reduces the coefficient associated with a linear regression of log hours worked per adult on log GDP per adult (column (1) of Table II). More specifically, the elasticity moves from -0.040 to -0.068 when weighting by population size, and from -0.045 to -0.056 when not weighting. All the other coefficients in the table remain almost identical. In particular, the main patterns by age and gender remain: hours worked by the young and the elderly decline with development, while hours worked by prime-age adults are flat.

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[1]More generally, all the other macroeconomic and policy variables used in our main cross-sectional analysis are for 2023 when available. When information for 2023 is not available, we use the latest year available.

Figure 1 visually depicts how using GDP and populations from the survey year instead of 2023 affects the overall relationship between hours per adult and development. Using survey year aggregates moves some countries slightly to the left of the figure, such as China and the Democratic Republic of the Congo, reflecting the fact that these countries were poorer at the time of the survey than they were by 2023. However, the overall picture remains the same. Hours worked are flat or slightly increasing with GDP per adult at lower levels of development; they then decline more sharply between upper-middle-income and high-income countries. There are considerable variations in hours worked at all levels of development. The adjusted R-squared associated with a regression of log hours worked on log GDP per adult is around 3-8% depending on the specification (weighting by population or not, using survey-year or 2023 aggregates).

2) Cross-Country and Within-Country Elasticities Gottlieb is also correct that in our panel database, which restricts the sample to 86 countries with long time series, hours worked are lower in richer countries (elasticity of -0.084, cross-country estimates), while economic growth is uncorrelated with changes in hours worked (elasticity of -0.012, within-country time series). The more negative cross-sectional elasticity is driven by the composition of the sample for which time series are available, as we mention in section V.E. of the paper:

"*As a starting point, we run a pooled panel regression of log hours on log GDP per adult without any fixed effects. This regression yields a more negative elasticity than in our cross-sectional database, between -0.1 (weighted) and -0.07 (unweighted). This result reflects the fact that our panel database covers fewer developing countries with low hours, especially Muslim/Hindu majority countries, and strongly overrepresents European countries with low hours.*"

More generally, on points 1) and 2), we do not view the estimation of the elasticity of hours per adult with respect to GDP as our fundamental contribution. We find sharp results on how global variations in working hours can be explained by basic demographic groups (young with schooling, elderly with pensions, gender reshuffling along intensive-extensive margins) and more suggestive evidence on tax policy, transfers, and regulations. How this aggregates up to an elasticity of total hours per adult with respect to GDP is less important to us. As our paper and this discussion shows, it is clear that there are enormous variations in hours worked across countries and over time that cannot be explained by development. As a result, there is not going to be a single elasticity parameter robust across cross-country datasets, panel datasets, choice of countries, weighting, etc.

3) Clustered Standard Errors and Coding Error in Country Groups Gottlieb also investigates the robustness of our main results on the panel database (Table II, Panel B and Table VI, Panel B) to clustering standard errors by country. With this specification, (1) the

small positive relationship between prime-age hours and GDP becomes non-significant and (2) the elasticities computed for subgroups of countries also lose statistical significance, except for the large positive elasticity of hours worked in Muslim/Hindu-majority countries.

We thank the author for this additional analysis and we agree that it is useful to report these estimates. Given the small samples of countries used in these subgroup analyses, it is natural that the estimates are noisier and should be interpreted as suggestive.

We are also grateful to the author for noticing a minor coding error in country groups. 41 country-years were indeed mistakenly coded as belonging to the Muslim/Hindu group. This affects the coefficient on "Other countries" in Table VI, which should be close to zero instead of negative.

5) Relationship between Hours per Adult, Employment Rates, and Hours per Worker Finally, Gottlieb also correctly points out that hours per adult are not exactly equal to hours per worker multiplied by employment rates for every single country-year.

The main reason for these small discrepancies is that the microdata samples used to measure hours per adult, hours per worker, and employment rates are sometimes not exactly identical. In particular, some individuals in the survey may have positive employment rates but missing hours worked, in which case we calculate employment rates over the entire sample but hours per adult over the restricted sample with non-missing hours (coding non-employed individuals as working zero hours, but employed individuals with missing hours as missing).

Fortunately, this inconsistency is rare and quantitatively small. The correlation between hours per adult and the product of employment rates and hours per worker is 0.9944 across approximately 2,500 surveys available in our panel database.

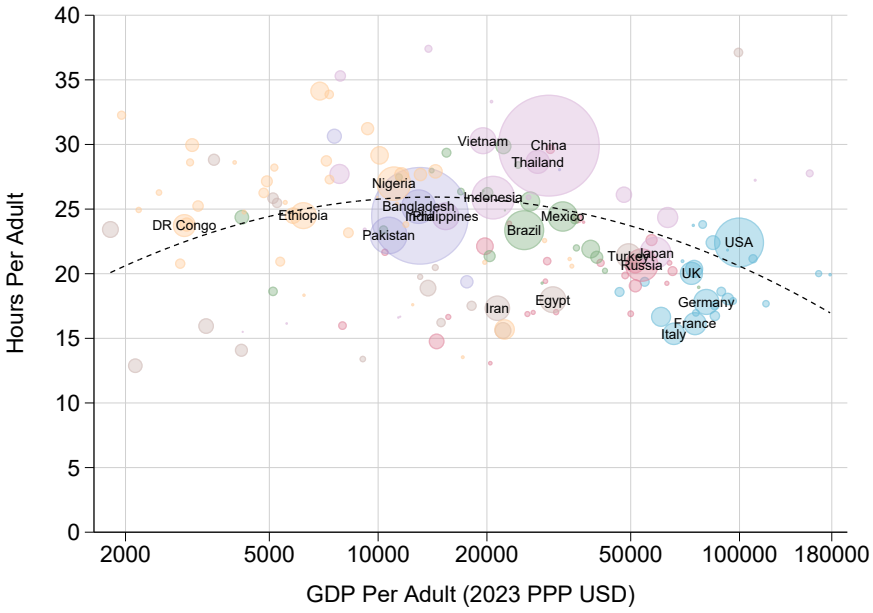
Nonetheless, while small in aggregate, this inconsistency can be quantitatively large for specific countries in specific years, such as Georgia and Uzbekistan circa 2000 as shown in Gottlieb (2026), Figure 2. We are grateful to the author for identifying this inconsistency and we will correct them systematically in future updates of the database.

References

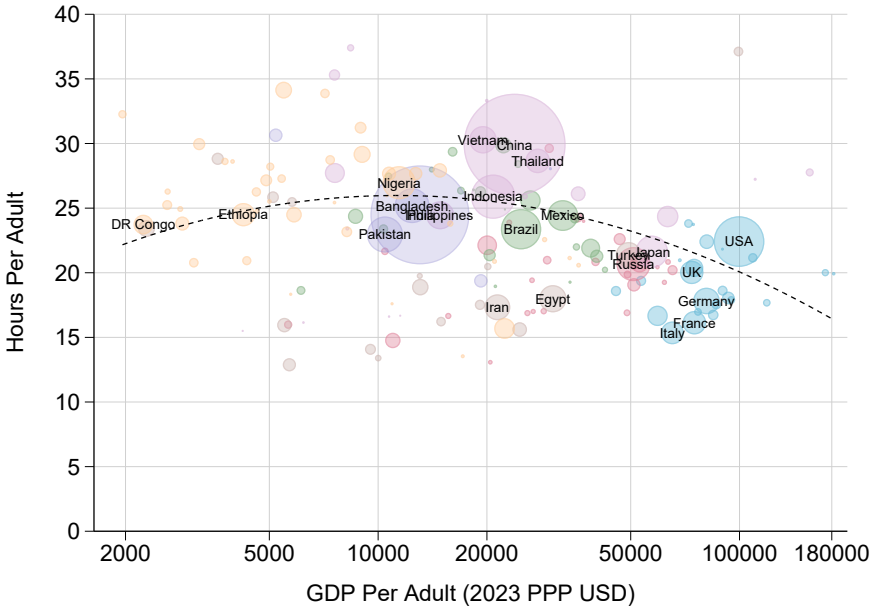
Gottlieb, C., "Hours Worked and Development: A Comment on Gethin and Saez (2026)," 2026.

Figure 1: Hours per Adult and Development: 2023 vs. Survey Years GDP and Populations

(a) GDP and Populations from 2023



(b) GDP and Populations from Survey Years



Notes: The figure shows average weekly hours of work per adult (aged 15+) against log GDP per adult in 2023 PPP US$. For each country, we use the most recent labor force survey available (generally 2022–2023 or 2019 as we exclude COVID years whenever possible). The area of each country's circle is proportional to its adult population; the largest countries' names are depicted. Colors correspond to world regions. The best quadratic fit of the weighted circles is represented by the dashed curve. Panel A uses 2023 estimates of GDP per adult and populations in each country. Panel B instead uses GDP and population estimates for the same year as the survey used to measure hours worked in each country.